

## 11.3 Graphing Exponential Functions Given an Equation

### Lesson Introduction

|                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                              |                                                                                                                                                                                           |                                                                                                                                                                                        |
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|  Benchmarks                                                     | MA.912.AR.5.3 Given a mathematical or real-world context, classify exponential functions as representing growth or decay.<br>MA.912.AR.5.6 Given a table, equation, or written description of an exponential function, graph that function and determine its key features.                                                                                                                                                                                                                                                                                                                       |                                                                                              |  Learning Targets                                                                                      | <ul style="list-style-type: none"><li>I can graph an exponential function given the equation.</li><li>I can classify a function as representing exponential growth or decay.</li></ul> |
|  Benchmark Coherence                                            | Building On                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                              | Working Toward                                                                                                                                                                            |                                                                                                                                                                                        |
|                                                                                                                                                  | N/A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                              | MA.912.AR.5.7 Solve and graph mathematical and real-world problems that are modeled with exponential functions. Interpret key features and determine constraints in terms of the context. |                                                                                                                                                                                        |
|  Essential Questions                                            | <ul style="list-style-type: none"><li>How can you graph an exponential function given the equation?</li><li>What is the relationship between the graph of an exponential function and its equation?</li></ul>                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                              |                                                                                                                                                                                           |                                                                                                                                                                                        |
|  Lesson Narrative                                               | In this lesson, students continue their learning of exponential functions and begin focusing on graphing an exponential function given the equation. Students will begin by recalling what they know about key features given an exponential function, then sketch a graph by using the equation to complete a table of values. They will collaborate with a partner to conceptualize the process for graphing exponential functions and determining their shapes and structures. Students will also complete a card sort activity, matching graphs with their key features and table of values. |                                                                                              |                                                                                                                                                                                           |                                                                                                                                                                                        |
|  Component Summary                                            | 11.3.1 Warm-Up (~5 min.)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Career exploration: registered nurse ( <i>SE p. 171</i> )                                    | 11.3.4 Activity (15 min.)                                                                                                                                                                 | Card sort activity where students match the graph, $y$ -intercept, $a$ and $b$ values, equation, and tables of values for different functions ( <i>SE p. 174</i> )                     |
|                                                                                                                                                  | 11.3.2 Guided Instruction (10-15 min.)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Formal instruction on graphing an exponential equation ( <i>SE p. 171</i> )                  | 11.3.5 Your Turn! (5-7 min.)                                                                                                                                                              | Independent practice graphing an exponential function and determining its key features ( <i>SE p. 174</i> )                                                                            |
|                                                                                                                                                  | 11.3.3 Collaboration (5-10 min.)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Collaborate to describe and sketch exponential graphs given an equation ( <i>SE p. 173</i> ) | 11.3.6 Wrap-Up (~5 min.)                                                                                                                                                                  | Ticket Out the Door graphing an exponential function ( <i>SE p. 175</i> )                                                                                                              |
|  Resources                                                    | Glossary                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                              | Materials                                                                                                                                                                                 |                                                                                                                                                                                        |
|                                                                                                                                                  | <u>Key Terms:</u> N/A<br><br><u>Additional Terms:</u> N/A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                              | <ul style="list-style-type: none"><li>Card sort activity cards</li></ul>                                                                                                                  |                                                                                                                                                                                        |
| <div>Additional Preparation</div> <ul style="list-style-type: none"><li>11.3.1 Warm-Up contains a video to accompany question prompts.</li></ul> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                              |                                                                                                                                                                                           |                                                                                                                                                                                        |

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|-----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <br><br>Teacher<br>Tips              | Common Misconceptions                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Things to Consider                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                                                                                                       | <ul style="list-style-type: none"><li>Students may make calculation errors when creating a table of values.</li><li>Students may start the graph at the y-intercept and not include values past the y-intercept.</li><li>Students may switch growth for decay or positive for negative in the function.</li></ul>                                                                                                                                                                                | <ul style="list-style-type: none"><li>Success in this lesson depends upon student knowledge from the prior lesson. Consider using data from the previous lesson to determine which students will need support determining key features and arrange to work with them during this lesson.</li><li>If time is an issue, consider assigning 11.3.5 Your Turn! for homework.</li></ul>                                                           |
|                                                                                                                       | Literacy and Language Routines                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Mathematical Thinking and Reasoning (MTR) Standard                                                                                                                                                                                                                                                                                                                                                                                           |
|                                                                                                                       | <b>Compare and Connect</b><br>11.3.2 Guided Instruction provides students with formal instruction on graphing an exponential equation. This can tie directly into what they learned about key features and growth/decay in this unit and graphing in prior units. Consider leveraging the Compare and Connect routine to provide students with a verbal version of your thinking so they can follow along with the rationale of why you are doing what you're doing with the function and graph. | <b>MA.K12.MTR.2.1: Multiple Representations</b><br>Throughout this lesson, students are continuing their learning with exponential functions, but are seeing them represented in multiple ways. In this lesson, they are provided the equation but will represent that same function using a table and a graph. Continuing to make connections between the different representations helps students build stronger conceptual understanding. |
| <br><br>Differentiate<br>Instruction | For 11.3.4 Activity, consider having students post their card sort activity answers in various locations around the room. Use these representations to have a discussion around how they know the cards match based on what they've learned.                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                                                                                                                       | 11.3.6 Wrap-Up provides students with an opportunity to reflect on their level of understanding of graphing an exponential function. Student response to this activity can provide valuable data on whether students have mastered the learning outcomes of this lesson.                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Lesson Notes:                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                              |

## 11.3.1 Warm-Up: Registered Nurse

**Component Overview:** Students will engage in a career exploration of a registered nurse and respond to two questions independently.



### 11.3.1 Warm-Up: Registered Nurse

1. Keith Koga is a registered nurse. He said he's had a lot of mentors, some of which included nurses he worked with. The culture of the workplace was one where everyone supported each other and knew they could go to each other for help.

- a. Who is someone you consider a mentor in your life?

*Answers will vary.*

- b. When learning something new, why is it helpful to ask someone (a peer, a teacher, a mentor, etc.) when you need help?

*Answers will vary.*



*For this activity, students will explore the career of a registered nurse. Prior to reading the narrative of this question, students should view the video (2:57 in length) and then read and respond to the questions.*

## 11.3.2 Guided Instruction: Graphing an Exponential Equation

**Component Overview:** Students will receive guided instruction on graphing an exponential function. They will first determine the key features of a given exponential function. Then, students will use the same function to create a table of values and sketch a graph using those values. This activity will conclude with students exploring the same function where the first value changes signs.



### 11.3.2 Guided Instruction: Graphing an Exponential Equation

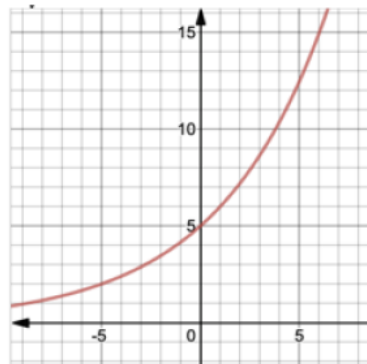
When graphing an exponential function, knowing the key features and using a table of values can be helpful.

1. Explain what key features can be determined from the equation  $C(x) = 5\left(\frac{6}{5}\right)^x$ .

| Key Feature                     | Explanation                                                                                                                                                                              |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Growth or Decay                 | <i>The value of <math>b</math> can be used to determine if it's growth (<math>b &gt; 1</math>) or decay (<math>0 &lt; b &lt; 1</math>).</i>                                              |
| Domain and Range                | <i>Domain: <math>\mathbb{R}</math> because it is exponential<br/>Range: <math>y &gt; 0</math> because <math>a &gt; 0</math></i>                                                          |
| Increasing / Decreasing         | <i>This is a positive exponential growth function, so it is always increasing.<br/><math>-\infty &lt; x &lt; \infty</math></i>                                                           |
| Constant Percent Rate of Change | <i>The value of <math>b</math> is <math>\frac{6}{5}</math>, giving a constant percent rate of change of 120%.</i>                                                                        |
| End Behavior                    | <i>Left: <math>x \rightarrow -\infty, y \rightarrow 0</math> Right: <math>x \rightarrow \infty, y \rightarrow \infty</math> because <math>a &gt; 0</math> and <math>b &gt; 1</math>.</i> |
| Positive / Negative             | <i>Positive: <math>-\infty &lt; x &lt; \infty</math> because <math>a &gt; 0</math>.</i>                                                                                                  |

2. Complete the table for  $C(x) = 5\left(\frac{6}{5}\right)^x$ . Then, graph the function.

| $x$ | $C(x)$           |
|-----|------------------|
| -2  | $\frac{125}{36}$ |
| -1  | $\frac{25}{6}$   |
| 0   | 5                |
| 1   | 6                |
| 2   | $\frac{36}{5}$   |
| 3   | $\frac{216}{25}$ |



This first question relies heavily on student mastery of the first two lessons in this unit. While the graphing included in this component should be guided, this question can be released to students and used as a gauge of their understanding or as a review of what they've learned. Make note of any students who are struggling with any of these specific key features to plan for how to provide support throughout the remainder of the unit.



- How can you determine that this function models growth and not decay?
- How does knowing whether the function models growth or decay help with determining the key features?

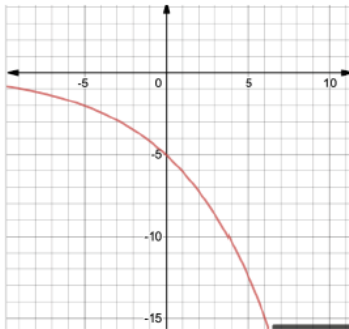


When providing instruction on question 2, consider explaining to students why a table of values is being created. Students should be familiar with this process from their prior units with graphing in both linear and quadratic situations. However, it will be important to conceptually connect students to the process.

## 11.3.2 Guided Instruction: Graphing an Exponential Equation (continued)

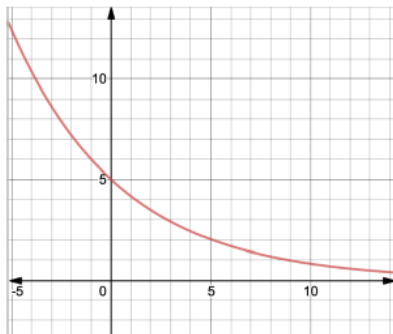
- a. Discuss with your partner how the graph would change if the  $a$ -value of 5 was changed to  $-5$ .  
*Answers will vary. The graph would reflect across the  $x$ -axis.*

- b. Draw a sketch of  $C(x) = -5\left(\frac{6}{5}\right)^x$ .



- c. Discuss with your partner how the graph would change if the  $b$ -value was changed from  $\frac{6}{5}$  to  $\frac{5}{6}$ .  
*Answers will vary. It would become a decay function causing a downward curve that approaches the  $x$ -axis.*

- d. Draw a sketch of  $C(x) = 5\left(\frac{5}{6}\right)^x$ .



- What are the benefits and limitations of a graph that is sketched?
- What is a way we could check our sketch graph against the actual graph?



For an extension to this activity, consider having students verify their sketched graph in question 2d to the actual graph. Provide students with a graphing utility in order to explore what this graph looks like compared to what they sketched.

## 11.3.3 Collaboration: Graphing Exponential Functions

**Component Overview:** Students will work with a partner to complete a series of statements about the key features of exponential functions, along with graphing the exponential function.



### 11.3.3 Collaboration: Graphing Exponential Functions

1. Consider the function  $f(x) = 3(1 - 0.32)^x$ .
- a. Work with your partner to complete the statements about  $f(x) = 3(1 - 0.32)^x$ .

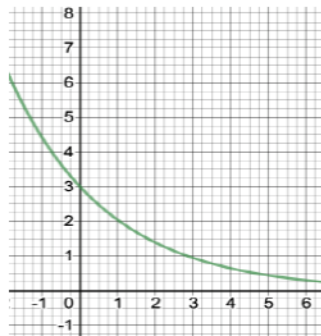
Since  $a$  is ☐ positive ☐ negative and  $b$  is ☐ between 0 and 1, ☐ greater than 1,

the function is ☐ increasing. ☐ decreasing. The left end behavior of

the graph is ☐  $y \rightarrow 0$  ☐  $y \rightarrow \infty$  and the right end behavior is

☐  $y \rightarrow 0$ . ☐  $y \rightarrow -\infty$ . The  $y$ -intercept of the function is ☐ 0.32. ☐ 0.68. ☐ 3.

- b. Sketch the graph of  $f(x) = 3(1 - 0.32)^x$ .



- What would happen if the  $a$  is negative and the  $b$  is between 0 and 1? How would this statement change?
- What would happen if the  $a$  remained positive but the  $b$  is greater than 1? How would this statement change?



When reviewing the answer to question 1a, consider leveraging an online graphing utility to show the graph in order to save time and show how the actual graph compares to a sketch.

## 11.3.3 Collaboration: Graphing Exponential Functions (continued)

2. Consider the function  $f(x) = -(1 + 0.4)^x$ .

a. Work with your partner to complete the statements for  $f(x) = -(1 + 0.4)^x$ .

Since  $a$  is ☐ positive ☒ negative and  $b$  is ☐ between 0 and 1, ☒ greater than 1,

the function is ☐ increasing. ☒ decreasing. The left end behavior of

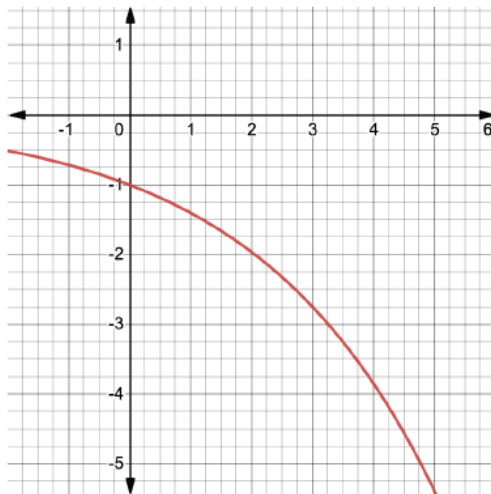
the graph is ☐  $y \rightarrow 0$  ☒  $y \rightarrow -\infty$ , and the right end behavior is

☐  $y \rightarrow 0$ . ☒  $y \rightarrow -\infty$ . The  $y$ -intercept of the function is ☐ 0.4. ☒ -1. ☐ 1.04.



- What is different about the exponential function represented in this question versus the one in the prior question?
- Why do you think the graph has a  $b$  value that signifies growth, but has a graph that doesn't increase or produce positive  $y$ -values as  $x \rightarrow \infty$ ?

b. Sketch the graph of  $f(x) = -(1 + 0.4)^x$ .



## 11.3.4 Activity: Card Sort

**Component Overview:** Students will complete a card sort activity where they match the graph with the  $y$ -intercept, the  $a$  and  $b$  values, the equation, and a table of values representing the same function.



### 11.3.4 Activity: Card Sort

Work with your partner to complete the card sort. Fill in the table with the correct card labels that correspond to each graph.

| Card Sort |                |                |           |            |            |
|-----------|----------------|----------------|-----------|------------|------------|
| Graph     | $y$ -Intercept | $a =$<br>$b =$ | Function  | Table      | Table      |
| 1.1       | <i>A1</i>      | <i>T1</i>      | <i>W1</i> | <i>Z1</i>  | <i>AC1</i> |
| 1.2       | <i>B1</i>      | <i>U1</i>      | <i>X1</i> | <i>AA1</i> | <i>AD1</i> |
| 1.3       | <i>A1</i>      | <i>D1</i>      | <i>G1</i> | <i>J1</i>  | <i>M1</i>  |
| 1.4       | <i>B1</i>      | <i>E1</i>      | <i>H1</i> | <i>K1</i>  | <i>N1</i>  |
| 1.5       | <i>A1</i>      | <i>F1</i>      | <i>I1</i> | <i>L1</i>  | <i>O1</i>  |
| 1.6       | <i>B1</i>      | <i>S1</i>      | <i>V1</i> | <i>Y1</i>  | <i>AB1</i> |



Consider grouping or pairing your students based on their performances in the prior lesson, as it is essential to success in this activity. Based on your groupings, determine which groups you may need to visit first to help ensure they are on the right track.

Consider the use of a timer or other method that indicates when your fixed time for this activity is completed.



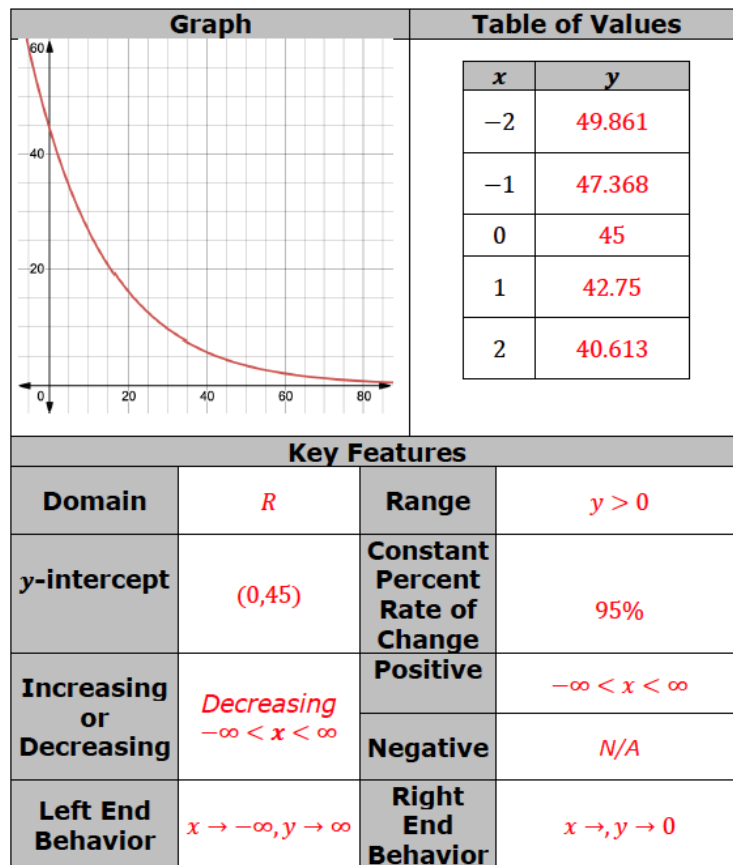
## 11.3.5 Your Turn!

**Component Overview:** Students will complete an independent practice activity where they will graph an exponential function and determine its key features.



### 11.3.5 Your Turn!

- Graph the exponential equation  $y = 45(1 - 0.05)^x$ . If necessary, complete the table of values. Then, determine all the key features.



For students that struggled in the card sort activity, this activity provides a great opportunity for one-on-one or small-group remediation and support. Consider helping students who struggled in the card sort activity with how to get started on this question and address the processes or key features that they are still confused on.

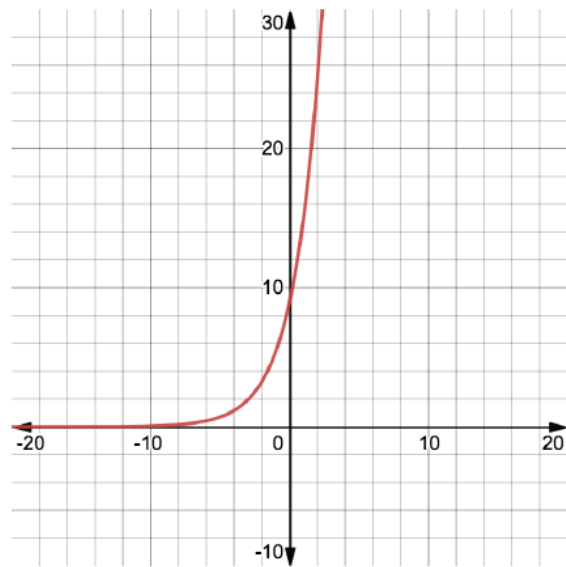
## 11.3.6 Wrap-Up: Ticket Out the Door

**Component Overview:** Students will complete a Ticket Out the Door that assesses student understanding of graphing an exponential function.



### 11.3.6 Wrap-Up: Ticket Out the Door

1. Graph the function  $y = 9\left(\frac{5}{3}\right)^x$ .



Student responses to this activity can provide valuable data on whether students have mastered the learning outcome of this lesson.